CAPSTONE DEFENCE

# Certificates a shipowner can verify on their own

Fund a bounded local pilot of sovereign evidence cells.
Do not fund Azure deployment yet.

**Green-Fuel Bunkering Certification Platform**

Phase 0: design, controls and evidence. No application and no Azure resource exists.



SAMPLE · NOT APPROVED FOR PRESENTATION

SAMPLE · NOT APPROVED FOR PRESENTATION

# A green-fuel premium is only paid if the certificate is believed

Each delivery must become a certificate that survives a shipowner, an auditor and a dispute.

1 **Meter readings** → 2 **Blend and feedstock** → 3 **Emissions attribute** → 4 **Certificate** → 5 **Shipowner and auditor**

## WHERE IT BREAKS

- Evidence sits in many places
- Methods change over time
- Disputes need proof

Sources PROBLEMSTATEMENT.md · STAKEHOLDERS.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# 8 stakeholders decide this, and the case names only 2

Whoever can fund it, approve a filing, open a meter or accept the evidence. None has been interviewed.

| STAKEHOLDER | BASIS | DECIDES |
|---|---|---|
| **CEO / group executive sponsor** | Candidate | Funding, risk appetite, whether to proceed |
| **Quality manager** | Stated | Approval of every certification filing… |
| **Trading head** | Candidate | Green-fuel pricing, customer commitments… |
| **Terminal operator** | Candidate | Access to meters, tanks, networks and staff… |
| **Barge operator** | Candidate | Edge equipment on the vessel… |
| **Shipowner** | Stated | What it accepts as evidence… |
| **Certification auditor / body** | Candidate | Whether evidence and reconstruction satisfy… |
| **Compliance team** | Candidate | Legal reading of residency, privacy… |

Stated: the case study says so. Candidate: a hypothesis to confirm with that stakeholder.

Sources STAKEHOLDERS.md · PROBLEMSTATEMENT.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Reliability decides how much of the premium is kept

The premium pool is the prize and trusted certificates protect it. How much is unknown: the business has supplied no figure.

**$39.7M**

premium revenue pool a year, year 3, base inputs

SCENARIO

**$666,792**

of it protected by the platform, base case

SCENARIO

**0**

inputs supplied by the business so far

NOT PROVIDED

Scenario figures are assumptions in a model, not facts. They show where value would come from.

REVIEW Evidence missing: Group delivery volumes, green-fuel premiums and dispute history

Sources BUSINESS-VALUE-MODEL.md · COMMERCIAL-VALUE.md

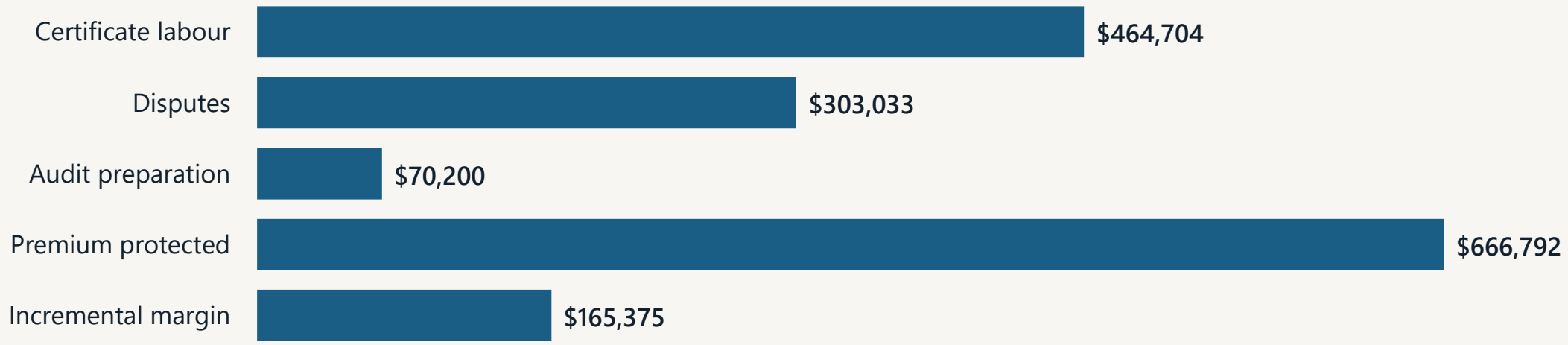
SAMPLE · NOT APPROVED FOR PRESENTATION

# Base case: real value, but no three-year payback

Placeholder inputs give a negative three-year NPV (-$1.56M) and about 60 months to pay back. Volume decides it.

**Annual benefit by source, year 3, base case**

SCENARIO




Break-even needs about 901 deliveries a year; the base case assumes 600. Doing nothing keeps the manual cost of about $985 per certificate and forgoes about $1.7M a year.

Sources BUSINESS-VALUE-MODEL.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Today's manual process is not known well enough, and fails where it is known

Records can be altered undetectably, effort grows with volume, and audit preparation is a document hunt.

| CRITERION | TODAY'S PROCESS |
|---|---|
| **Data residency** | where today's data sits is not known |
| **Evidence integrity** | document-based records can be altered undetectably |
| **Scalability** | effort grows with volume |
| **Cost** | no baseline cost exists (D-034) |
| **Operational complexity** | manual effort and error |
| **Security** | weak controls on paper records |
| **Certification auditability** | audit preparation is a document hunt |
| **Vendor dependency** | none, and none of the benefit |

The assessment is from the options analysis. The current systems themselves are not known (D-019, D-035).

Sources 02-ARCHITECTURE-OPTIONS.md · PROBLEMSTATEMENT.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Four options on eleven criteria: only sovereign cells can be made to meet residency

The cheapest option cannot show port residency. The recommended one pays for it in operational complexity.

| | A CENTRALISED | B EDGE PLUS SHORE | C SOVEREIGN CELLS | D TODAY |
|---|---|---|---|---|
| Data residency | Does not meet | Conditional | Conditional | Unknown |
| Evidence integrity | Conditional | Conditional | Conditional | Does not meet |
| Availability | Conditional | Conditional | Conditional | Unknown |
| Scalability | Meets | Conditional | Conditional | Does not meet |
| Performance | Conditional | Meets | Meets | Unknown |
| Cost | Meets | Unknown | Conditional | Unknown |
| Operational complexity | Meets | Does not meet | Does not meet | Does not meet |
| Security | Conditional | Conditional | Conditional | Does not meet |
| Certification auditability | Conditional | Conditional | Meets | Does not meet |
| Recovery | Conditional | Does not meet | Conditional | Unknown |
| Vendor dependency | Conditional | Conditional | Conditional | Meets |

Verdicts are the options analysis's own. Costs over three years: A $2.7M, B $3.8M (not like-for-like), C $5.2M.

ADR-0016 Blocked

Sources 02-ARCHITECTURE-OPTIONS.md · results.json

SAMPLE · NOT APPROVED FOR PRESENTATION

# Recommend sovereign cells, a deterministic core and bounded agents

Evidence stays in its port. Software computes and signs. AI drafts and explains, and never decides.

**Group plane** signed methodology and policy go down; only approved-flow metadata comes up (undecided, D-011)

SOVEREIGN CELL · PORT JURISDICTION A

**Edge buffer**
capture, sign, queue

→

**Verify and receipt**
duplicates ignored

→

**Evidence ledger**
append-only

**Deterministic services: software computes, signs, enforces**
calculation under a versioned method · certificate signing · approvals · audit

**Bounded agents (optional)**
draft and explain; never approve

**Per-shipowner access**
isolation enforced by the cell

CELL · JURISDICTION B

Same contract; hosting may differ: an Azure region or a local host, decided per port.

**One cell per jurisdiction, not per port.**

**Shipowner**
verifies offline, public key only

**Certification body**
only after quality-manager approval

Design, not built. The group plane carries no raw or evidence-grade data.

ADR-0016 Blocked | ADR-0017 Blocked | ADR-0018 Blocked | ADR-0019 Blocked

Sources 03-ARCHITECTURE-VIEWS.md · DES-0003-target-system-architecture.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# It wins on the hard constraints and pays for it in complexity

Five decisive reasons, in order. Each ties to a stated requirement or constraint.

| REASON | WHAT IT MEANS |
|---|---|
| **Residency by construction rather than by region…** | In C the unit of placement is the jurisdiction, so every store, backup, log stream… |
| **The system of record is not at the edge** | B puts the only durable copy on a small device. C treats the edge as a buffer… |
| **Agents stay possible** | SEC-003 allows AI inference over delivery data only in its jurisdiction… |
| **Blast radius and auditability** | A compromise, outage or audit touches one jurisdiction… |
| **It degrades gracefully per port** | A port with a verified in-country region runs an Azure cell… |

Conditions attached: if the sponsor will not pay for sovereignty, the choice is fewer ports, not a cheaper design that breaks residency.

ADR-0016 Blocked

Sources 02-ARCHITECTURE-OPTIONS.md · 08-DESIGN-DEFENSE.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Agents draft. Services validate and issue. A person approves

Five business agents work inside fixed boundaries; none can approve, file or change a method.

1
**Transfer completed**
signed events

→

2
**Evidence bound**
chain verified

→

3
**Emissions attribute**
deterministic code

→

4
**Certificate drafted**
by an agent

→

5
**Approved and issued**
quality manager, signed

→

6
**Shipowner verifies**
public key only

## THE RULES THAT HOLD

Agents never hold the permission to approve or file

Emissions come from versioned methods, never from a model

Every step ends in a defined, auditable state

Sources DEMO-EXECUTIVE.md · AGENT-AUTONOMY-MATRIX.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Evidence stays in its port: everywhere else is conditional or prohibited

A region name is not residency proof. Placement is proposed per data class and zone, and undecided means no.

| | Z1 EDGE | Z2 PORT SHORE | Z3 GROUP HUB | Z4 CLOUD/AI | Z5 CUSTOMER | Z6 STAFF ABROAD | Z7 EXTERNAL |
|---|---|---|---|---|---|---|---|
| Raw delivery measurement data | Yes | Yes | No | No | Cond | No | Cond |
| Delivery evidence packages | Yes | Yes | No | No | Cond | No | Cond |
| Blend and composition data | Yes | Yes | No | No | Cond | No | Cond |
| Feedstock certificates | Yes | Yes | TBD | No | TBD | No | Cond |
| Methodology and reference data | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Calculation records | Yes | Yes | Cond | No | Cond | No | Cond |
| Mass-balance ledger | Yes | Yes | Cond | No | No | No | Cond |
| Certificates | Yes | Yes | Cond | No | Cond | No | Cond |

Proposed rules. Yes permitted, Cond permitted only after a named decision, No prohibited, TBD undecided and treated as No.

ADR-0004 Blocked

ADR-0015 Blocked

Sources SOVEREIGNTY-MATRIX.md · 12-DATA-MOVEMENT-AND-RESIDENCY-VALIDATION.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Watch a delivery become a certificate a shipowner can verify

Two minutes. Real code on synthetic data. Every step is labelled, and a recording is announced as a recording.




Run: node harness/demo/run.mjs exec. Backup: 1 recorded run(s), each marked "notLive".

REVIEW Evidence missing: Measured reduction in certificate processing time

Sources DEMO-EXECUTIVE.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Measured: the rules do the work. AI value is unproven

On 42 synthetic cases a conventional workflow completed 41. No language model has been run.

**41 of 42**

synthetic cases completed, no model

MEASURED

**15 of 15**

emissions calculations correct

MEASURED

**0**

language-model arms run

NOT RUN

Synthetic, author-written, unreviewed; the holdout is not blind. No throughput, latency or availability test exists: there is no system to test.

Sources COMPARISON.md · AGENT-EVALUATION-REPORT.md

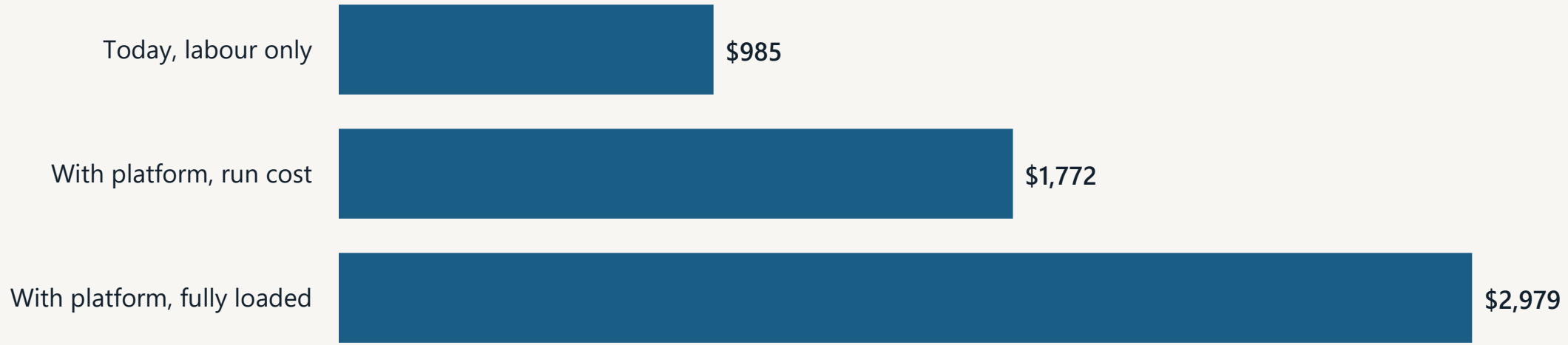
SAMPLE · NOT APPROVED FOR PRESENTATION

# Base case: $5.2M over three years, and more per certificate than today

Cost per certificate rises. The case rests on disputes and premium protection, not on labour.

**Cost per certificate, base case**

SCENARIO




One-time $2.3M; annual operating $985,308. 12 verified inputs (Azure list prices) and 79 scenario assumptions.

Sources BUSINESS-VALUE-MODEL.md · azure-prices.json

SAMPLE · NOT APPROVED FOR PRESENTATION

# Decision: fund a bounded local pilot. Not Azure deployment yet

Approve Phase 1a on synthetic data, and supply four inputs that turn assumptions into numbers.

| THE ASK | WHAT IT UNLOCKS |
|---|---|
| **Approve Phase 1a: local, synthetic, no Azure. Enrol an approver** | The first vertical slice |
| **Real volumes, premiums and dispute history** | A business case built on facts |
| **Name the ports. Counsel's reading on residency** | A hosting model per port |
| **A domain reviewer. Decide whether to test an LLM** | A reviewed evaluation |

Revisit the whole case if confirmed volume is well below about 901 deliveries a year.

Sources PHASE-1-BLOCKER-REPORT.md · TODO.md

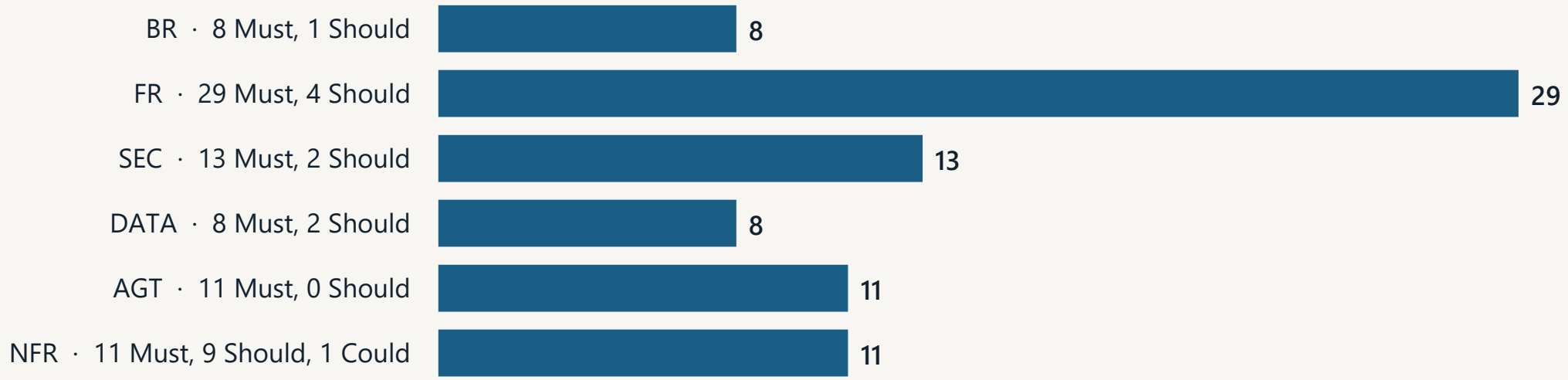
SAMPLE · NOT APPROVED FOR PRESENTATION

# 99 requirements, 80 of them Must, in six families

Case-study requirements are kept apart from assumptions, open questions and claims that still need verification.

## Must requirements by family

CASE STUDY




Counts are read from the requirements baseline; a Must requirement without a decision is listed in the appendix.

Sources REQUIREMENTS.md · REQUIREMENTS-TRACEABILITY.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Azure services are provisional leans, each with a residency check still to run

Selected by requirement, not by name. Alternatives were considered, and the open question per service is stated.

| ROLE | CANDIDATE | DISPOSITION |
|---|---|---|
| **Services and agent runtime** | Azure Container Apps | Provisional lean |
| **Ledger and metadata** | Azure Database for PostgreSQL | Provisional lean |
| **Evidence and certificates** | Azure Blob Storage | Provisional lean |
| **Ingestion queue** | Azure Service Bus | Provisional lean |
| **Signing keys** | Azure Key Vault Managed HSM | Provisional lean |
| **Tamper-evidence anchor** | Azure Confidential Ledger | Candidate, undecided |
| **Agent models** | Microsoft Foundry model deployments (Azure OpenAI models) | Deferred |
| **Identity** | Microsoft Entra ID | Provisional lean |

16 roles are assessed in the service decision matrix. None is verified in a region and none is deployed.

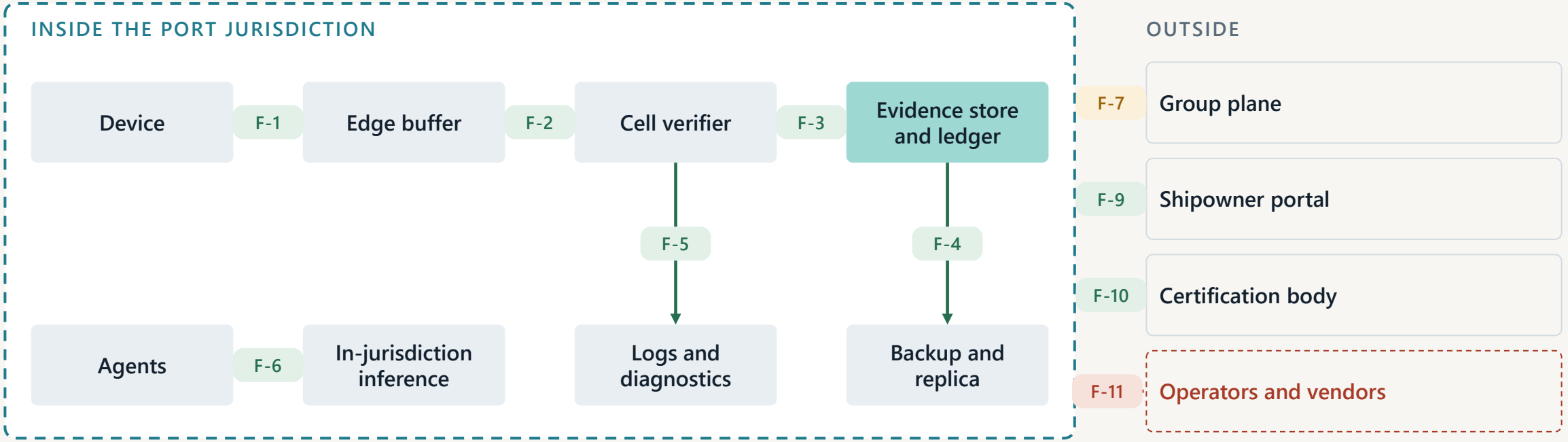
ADR-0019 Blocked

Sources 07-AZURE-SERVICE-DECISION-MATRIX.md · AZURE-REFERENCE-ARCHITECTURE.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Thirteen flows are classified. Only some may leave the jurisdiction

Edge links may be intermittent: events are buffered, signed and retried idempotently. Retries are duplicates, never new evidence.




Arrow colour: green allowed inside the jurisdiction, amber undecided, red forbidden by default. All 13 flows are in the appendix register.

ADR-0004 Blocked

Sources 12-DATA-MOVEMENT-AND-RESIDENCY-VALIDATION.md · 03-ARCHITECTURE-VIEWS.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Four agents may reach L4. One stays at L3. None can approve or file

L4 is a ceiling on autonomy inside a fixed boundary, not a permission to act commercially, regulatorily or on infrastructure.

| | L0 | L1 | L2 | L3 | L4 | L5 | WHY THIS CEILING |
|---|---|---|---|---|---|---|---|
| **Delivery Evidence (DEA)** | | | | | | | Ingests and binds source evidence as received within a fixed catalogue |
| **Emissions Attribute (EAA)** | | | | | | | Runs approved deterministic calculations… |
| **Dispute Resolution (DRA)** | | | | | | | Assembles packs; every outward or commercial step is human |
| **Mass-Balance (MBA)** | | | | | | | Posts events derived from signed source events… |
| **Customer Compliance (CCA)** | | | | | | | Answers in-policy questions directly to customers, with sampled review |

L0 No AI · L1 Assistive · L2 Drafting · L3 Conditional action · L4 Bounded autonomy · L5 Unbounded autonomy

Proposed ceilings. Approval, filing, method change and infrastructure change are never agent actions.

ADR-0017 Blocked

Sources AGENT-AUTONOMY-MATRIX.md · AZURE-REFERENCE-ARCHITECTURE.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# 59 controls in seven families. 9 cannot be tested until a product exists

Shipowner isolation is enforced per tenant by the cell, not by the agent. Each control names the requirement it serves.

## Controls by family

CANDIDATE




Security review: three HIGH, eleven MEDIUM and two LOW. Candidate oracles test the rules on synthetic data; no control is implemented in a product.

ADR-0017 Blocked

Sources 10-CONTROL-MATRIX.md · 13-SECURITY-REVIEW.md · 06-THREAT-MODEL.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Evidence is a signed chain; the certificate binds to its hash

A verifier needs only the certificate, the evidence files and a trust set it supplies. Tampering is detected, not prevented.

1
**Device signs each event**
key held by the device

2
**Cell verifies and receipts**
signed receipt, own clock

3
**Events chain by hash**
each hashes the previous

4
**Manifest binds evidence**
one hash for the transfer

5
**Certificate binds manifest**
quality manager signs

6
**Shipowner verifies**
public key, offline

**Tampering is detected, not prevented.** Change any event, receipt, manifest or certificate and a signature or hash no longer matches. An independent external anchor is an open decision (ADR 0018).

Recorded run (not live): 5 signed events, chain verified, 2,500 kg delivered, 5 receipts, tamper attempt detected.

ADR-0018 Blocked

Sources custody-chain-contract.mjs · certificate-lifecycle-contract.mjs · tamper-20261029T131741-5b3008eb.json

SAMPLE · NOT APPROVED FOR PRESENTATION

# Scalability is argued, not measured

The argument is structural: each cell scales independently. No throughput, latency or scale test has been run, because there is no system to test.

REVIEW PLACEHOLDER

EVIDENCE NEEDED

## Measured throughput, latency and scaling under load, per cell.

WHAT THE REPOSITORY HOLDS INSTEAD

- Placeholders and scenario arithmetic only; no target volume or response time exists (D-031, NFR-014, NFR-015).
- PH-C1: Deliveries per cell per year
- PH-C2: Peak concurrent deliveries per port
- PH-C3: Meters per delivery and reading rate
- PH-C4: Ingestion rate at peak

REVIEW Evidence missing: Measured throughput, latency and scaling evidence

Sources 05-QUALITY-ATTRIBUTES.md · COST-AND-CAPACITY.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Recovery stays inside each jurisdiction. Targets are unset and untested

Under failure the system refuses, waits or escalates. It never guesses. Regional outage and recovery are designed, not tested.

| LAYER | BASELINE DESIGN | STILL OPEN |
|---|---|---|
| **Evidence blobs** | Locally or zone redundant only… | Whether a second region in the same… |
| **Ledger database** | High availability inside the region as… | Recovery targets (D-031)… |
| **Messaging** | Zone redundancy where the tier supports it… | Replay rules after a failure… |
| **Keys and certificates** | Key custody in the jurisdiction (HC-03)… | HSM choice and ownership (D-013) |
| **Edge** | Store-and-forward with a local append-only… | Maximum disconnection and permitted… |
| **Control plane and IaC** | The repository and reviewed templates… | Where the pipeline runs and what it can see |
| **Customer-visible service** | Customer compliance answers can be degraded… | The degraded-mode behaviour per agent |

Edge outage, tamper, replay and injection are tested on candidate contracts. Region outage is not.

REVIEW Evidence missing: Tested recovery against agreed RPO and RTO

Sources OPERATIONS-AND-DR.md · 08-DESIGN-DEFENSE.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Observability is designed per cell. Nothing is instrumented yet

Logs and metrics stay with the data they describe. Telemetry is classified so it cannot leak what the cell must not export.

| NEED | DESIGN | NOT YET DECIDED OR VERIFIED |
|---|---|---|
| **Logs and metrics stay with the…** | One workspace per jurisdiction cell… | That the chosen region meets D-009 for logs… |
| **No sensitive data in telemetry** | Telemetry is classified; names of customers or vessels… | A logging standard for the application… |
| **Application and dependency health** | Application Insights workspace-based… | Instrumentation: no application exists |
| **Evidence-pipeline health** | Alerts on ingestion lag, dead-letter growth… | Thresholds: no target is agreed (D-031) |
| **Edge health** | Edge reports its last synchronisation… | The edge platform (D-018) |
| **Audit** | Every approval, every agent action and every denial is… | The audit store design |

A recorded observability demonstration exists (not live). No application emits telemetry.

REVIEW Evidence missing: Operational traces, dashboards and alert history from a running system

Sources OPERATIONS-AND-DR.md · observability-20261029T131742-b48a4268.json

SAMPLE · NOT APPROVED FOR PRESENTATION

# Four arms were scored. Both language-model arms are not run

The decision rule is fixed first: use rules by default and admit a model only where it beats the measured baseline.

| ARM | RESULT | STATE |
|---|---|---|
| **Conventional workflow (no model)** | 41 of 42 complete; all 9 hard gates pass | MEASURED |
| **Four deliberately bad arms** | All fail the gates, as they must (scorer check) | MEASURED |
| **Single-agent language model** | No run | NOT RUN |
| **Multi-agent language model** | No run | NOT RUN |

Synthetic, author-written, unreviewed cases; the holdout is not blind. Model selection is deferred (D-028).

Sources COMPARISON.md · AGENT-EVALUATION-REPORT.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# What was rejected, and the requirement that rejected it

Every rejection is stated in terms of a requirement or constraint, not preference.

| ALTERNATIVE | DISPOSITION | REASON |
|---|---|---|
| **A Centralised Azure** | Rejected (hard constraint) | Violates HC-01 for any group with more than one port jurisdiction… |
| **B Edge plus shore** | Not selected | Not a hard-constraint violation if the shore platform is confined to derived data, but it… |
| **D Today** | Not selected | Does not meet HC-08 (undetectable alteration), does not scale with volume… |
| **Azure Functions** | Not selected | Adds a second programming and operations model for little gain at this volume… |
| **App Service** | Not selected | No requirement that Container Apps cannot meet; one fewer service to verify |

What would change the recommendation is listed in the options analysis.

ADR-0016 Blocked

Sources 02-ARCHITECTURE-OPTIONS.md · 07-AZURE-SERVICE-DECISION-MATRIX.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# 6 critical gaps stand between this repository and a defensible capstone

They are stated before anyone asks. Each says how it is closed and by whom.

| GAP | WHAT IS MISSING | CLOSED BY |
|---|---|---|
| **G-01** | No working solution end to end. No src/, no business agent, no Azure resource… | build, owner |
| **G-02** | No AI solution and no model evaluation of any business agent… | build |
| **G-03** | No numbers the audience can act on. No latency, throughput, cost per delivery… | build, owner |
| **G-04** | No decided solution architecture and no recommendation… | owner, build |
| **G-05** | Thought-leadership artifact not chosen, finished or submitted… | owner, publish |
| **G-06** | No narrative, deck, 2-minute demo or rehearsal material. Section B… | build, live |

From the forensic audit at the commit it names; gaps closed since are not re-marked here.

Sources CRITICAL-GAPS.md · RUBRIC-TRACEABILITY.md

# The path to production runs through nine gates. None is passed

Production deployment is never automatic: each gate needs a person, and Azure deployment is deferred by the owner.




Gate G1 (phase raised) and G2 (approver enrolled) come first. Architecture baseline authorization is built; no baseline exists.

Sources PHASE-1-BLOCKER-REPORT.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# The target architecture is Blocked in the register, not approved

Status is read from the decision register at this commit, never written into the deck.

| ADR | DECISION | STATUS | APPROVAL BASIS |
|---|---|---|---|
| **ADR-0004** | Hold delivery measurements in jurisdiction-local authoritative stores | Blocked | none |
| **ADR-0015** | Jurisdiction-local, tamper-evident evidence ingestion by an edge… | Blocked | none |
| **ADR-0016** | Sovereign evidence cells per jurisdiction with federated coordination | Blocked | none |
| **ADR-0017** | Deterministic core and bounded business agents | Blocked | none |
| **ADR-0018** | Evidence integrity and certificate mechanism… | Blocked | none |
| **ADR-0019** | Azure services are selected by requirement and verified per… | Blocked | none |
| **ADR-0020** | Phase 1a, a local synthetic-data vertical slice inside this repository | Proposed | none |

13 of 58 register decisions have the status Approved; 12 hold an approval basis the policy accepts. 46 are not yet decided.

ADR-0004 Blocked | ADR-0015 Blocked | ADR-0016 Blocked | ADR-0017 Blocked

Sources 0016-sovereign-evidence-cells-with-federated-coordination.md · 0020-phase-1a-local-synthetic-vertical-slice.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# 3 Must requirements have no decision yet

Each requirement traces to the decisions it depends on, with its acceptance criteria and evidence required.

| FAMILY | REQUIREMENTS | MUST | TRACED TO A DECISION |
|---|---|---|---|
| **BR** | 9 | 8 | 8 |
| **FR** | 33 | 29 | 33 |
| **SEC** | 15 | 13 | 14 |
| **DATA** | 10 | 8 | 8 |
| **AGT** | 11 | 11 | 9 |
| **NFR** | 21 | 11 | 19 |

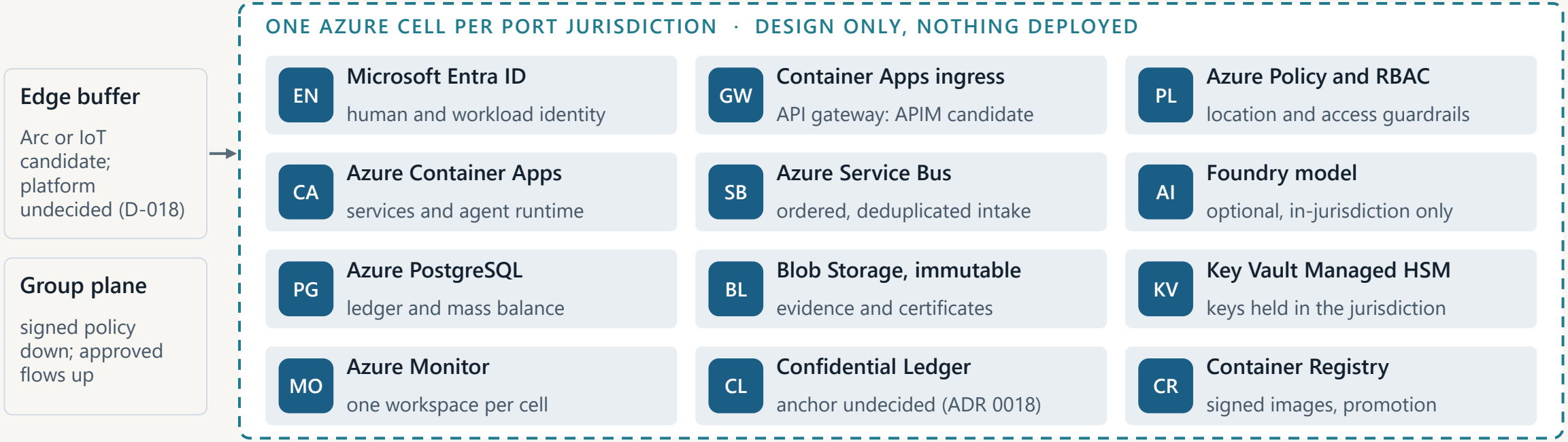
Must requirements without a decision: BR-009, DATA-003, AGT-009.

Sources REQUIREMENTS-TRACEABILITY.md · REQUIREMENTS-COVERAGE.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Reference topology: one Azure cell per port jurisdiction, design only

No Azure resource has been created. The icons are placeholders until the official icon set is supplied.




Provisional leans from the service decision matrix. Region, quota and residency are unverified for every service.

ADR-0019 Blocked

Sources AZURE-REFERENCE-ARCHITECTURE.md · 07-AZURE-SERVICE-DECISION-MATRIX.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# The 13 classified data flows, with the design position of each

Only some may leave the jurisdiction. Validation of each flow against a real network path is still to do.

| FLOW | FROM TO | DESIGN POSITION |
|---|---|---|
| **F-1** | Device to edge buffer | Allowed, inside the jurisdiction, store-and-forward |
| **F-2** | Edge buffer to cell verifier | Allowed, inside the jurisdiction |
| **F-3** | Cell verifier to cell evidence store and ledger | Allowed, inside the jurisdiction |
| **F-4** | Cell to cell backup and replica | Allowed only to a target inside the jurisdiction |
| **F-5** | Cell to cell log and diagnostic store | Allowed, inside the jurisdiction; export only on an approved flow |
| **F-6** | Cell agents to the inference endpoint | Allowed only to an endpoint in the jurisdiction whose processing location is… |
| **F-7** | Cell to group plane | Undecided. Each item needs a class and an approved-flow entry with a legal basis |
| **F-8** | Group plane to cell | Allowed; signed, verified at the cell |
| **F-9** | Cell to shipowner portal | Allowed; per-tenant |
| **F-10** | Cell to certification body | Allowed only after quality-manager approval, via the controlled submitter |
| **F-11** | Any operator or vendor path into a cell from outside | Forbidden by default; break-glass approved inside the jurisdiction |
| **F-12** | Deployment pipeline to a cell | Allowed through scoped deployment identities; must not be a way to read cell data |
| **F-13** | Real data to any non-production environment | Forbidden; synthetic or approved-masked data only |

ADR-0004 Blocked

Sources 12-DATA-MOVEMENT-AND-RESIDENCY-VALIDATION.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# There is no performance evidence yet

This slide is a placeholder by design: it is replaced when a running slice is measured.

REVIEW PLACEHOLDER

EVIDENCE NEEDED

**Throughput, latency, scale and availability measurements with method, environment, and raw results.**

WHAT THE REPOSITORY HOLDS INSTEAD

- No application and no Azure resource exist.
- Capacity method exists; every figure is withheld (see cost and capacity).

REVIEW Evidence missing: Performance results

Sources COST-AND-CAPACITY.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Cost: the method exists; the Azure run cost has no figure

79 scenario assumptions, 12 verified inputs, 1 benchmark, 0 supplied by the business.

| ITEM | WHY THERE IS NO FIGURE |
|---|---|
| **Monthly run cost per jurisdiction cell** | No SKU, volume, retention, redundancy or region is decided |
| **Cost of the edge or on-premises evidence store** | The edge platform and hardware ownership are open (D-018, D-010) |
| **Cost of model use** | The models, deployment types and prompt volumes are open (D-028) |
| **Non-production cost** | The number and scale of environments is open (D-037) |
| **Budget** | The cost envelope is open (D-034) |
| **Capacity and quotas** | No target volume or response time exists (D-031, NFR-014, NFR-015)… |

The business value model uses Azure list prices only as verified inputs; its totals are scenario results.

Sources COST-AND-CAPACITY.md · BUSINESS-VALUE-MODEL.md

# 34 threats across six trust boundaries, each with a control and a stated limit

The security review found three HIGH findings; none is closed by a product because none exists.

**34**

threats in the threat model

DESIGN

**59**

controls mapped

CANDIDATE

**9**

controls not yet testable

NOT BUILT

Security review: three HIGH, eleven MEDIUM and two LOW. Boundaries: device to edge, edge to cell, agents to tools, shipowner to portal, group plane to cell, certificate to outside.

Sources 06-THREAT-MODEL.md · 13-SECURITY-REVIEW.md · 10-CONTROL-MATRIX.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# 41 of 42 synthetic cases complete; the one miss is a free-text relative date

Hard gates passed. Language-model arms did not run.

**97.6%**

task completion rate

MEASURED

**15 of 15**

emissions calculations correct

MEASURED

**9 of 9**

hard gates passed

MEASURED

Run AR-deterministic-workflow-3f7a0688. Synthetic cases written by the author and not reviewed; the holdout is not blind.

Sources COMPARISON.md · AGENT-EVALUATION-REPORT.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# 1,056 of 1,056 automated tests pass on the recorded run

They test the harness and candidate contracts on synthetic data. They do not show a requirement is satisfied.

**1,056**

tests passing, recorded run EV-0077

MEASURED

**0**

application tests

NOT BUILT

**0**

end-to-end tests of the platform

NOT BUILT

Requirement coverage is generated and states what a test that ran has shown.

Sources EV-0077.json · REQUIREMENTS-COVERAGE.md

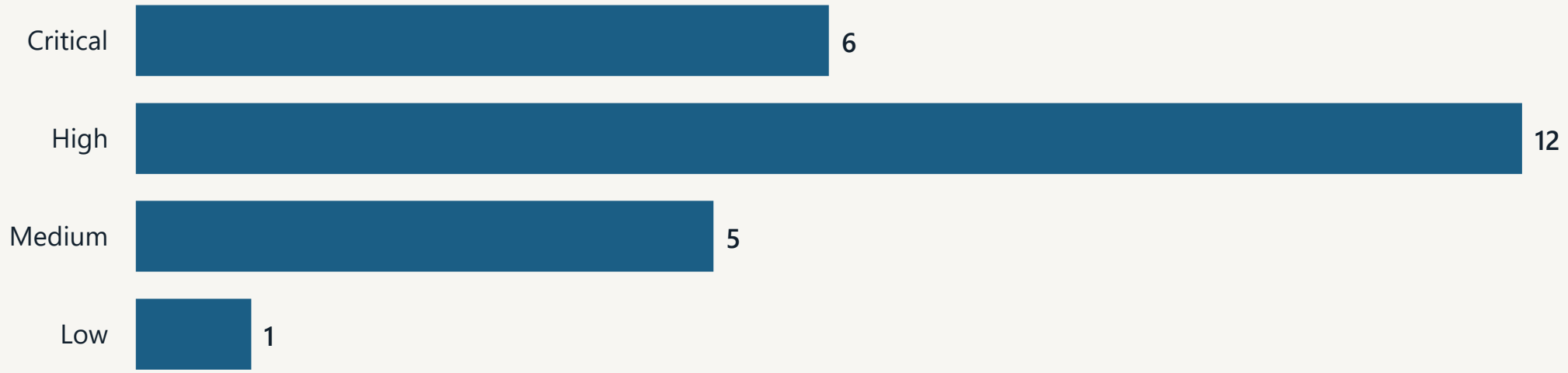
SAMPLE · NOT APPROVED FOR PRESENTATION

# 24 gaps were recorded at the audit. 6 are critical

The register is dated to its audit commit. A gap closed since is not marked closed here.

## Gaps by severity at the audit

DESIGN




Rubric floors: no criterion scored 1; each section at least 60 percent; artifact submitted.

Sources CRITICAL-GAPS.md

SAMPLE · NOT APPROVED FOR PRESENTATION

# Fifty panel questions are prepared. One per area is shown here

Each has a recommended answer, its evidence, what is uncertain, and when to concede.

| ID | AREA | QUESTION |
|---|---|---|
| **Q01** | Discovery | Have you actually spoken to a bunker supplier, or is this a desk exercise? |
| **Q02** | Alternatives | Why not use spreadsheets, an ERP module or an existing certification registry? |
| **Q03** | Commercial | Where does the premium revenue figure come from? |
| **Q14** | Architecture | One cell per port looks over-engineered. Defend it. |
| **Q15** | AI governance | The brief says L4 agents. What stops an agent doing harm? |
| **Q18** | Delivery | You built a harness and no product. Is this over-engineered for a capstone? |
| **Q21** | Demo | Was that live or recorded? |
| **Q24** | Disaster recovery | What happens if the Azure region goes down? |
| **Q26** | Security | What if a signing key is stolen? |

Full answers: docs/capstone/presentation/QA-PANEL-50.md. Delivery under challenge is not verified until a human rehearsal.

Sources QA-PANEL-50.md

# Presentation criteria: what the deck can serve and what only a live session can

No official score is assigned. Delivery, listening and composure are not verified until a person has rehearsed.

| CRITERION | EVIDENCE NOW | SLIDES | HOW ASSESSED |
|---|---|---|---|
| **B1 Narrative structure & argument** | None | E01, E02, E02B, E05 | L, H |
| **B2 Audience calibration & register** | Thin | T02, T03, T04, T05 | L, H |
| **B3 Demo & executive hook** | Thin | E09, E11 | L |
| **B4 Q&A composure & depth under…** | Partial | T02, T03, T04, T05 | L, H |
| **B5 Delivery craft: pace…** | None | live only | L |
| **B6 Written communication…** | Partial | E07, T01, A01, A02 | T, H |
| **B7 Use of evidence & data…** | Thin | E03, E04, E06, E08 | T, L |
| **C1 Discovery & business framing** | Documented | E02, E02B, E03, E04 | L, H |
| **C2 Active listening & disagreement…** | None | A10 | L |
| **C3 Influence & intellectual honesty** | Demonstrated honesty | E01, E03, E04, E12 | L, H |

Mode: T technical evidence, L scored live, H human judgment, P publication.

REVIEW Evidence missing: Human rehearsal of the B and C criteria

Sources RUBRIC-TRACEABILITY.md · REHEARSAL-PLAN-AND-SCORECARD.md

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# This deck was generated from one commit and a recorded set of sources

A slide is only as good as the evidence it cites. The generation record beside the file lists every source with its hash.

| ITEM | VALUE |
| --- | --- |
| **Repository commit** | dff92840d64e628bb2ce33fe97f40d72f2bce9dd |
| **Branch** | studio/presentation-studio |
| **Slides** | 41 |
| **Sources read** | 36 files, digest a9b10a7748338e32 |
| **Claims checked** | 40 on the other slides; 0 unsupported |
| **Rubric** | derived traceability; the official PDF is not in the repository |

Sources RUBRIC-TRACEABILITY.md